

Clause-Activation Signatures (CAS) for Attack-Family Attribution in IoMT

Rakesh Reddy Yakkati, Per-Arne Andersen, Lei Jiao, Ole-Christoffer Granmo, and Rishad Shafik

Abstract—When an intrusion detector flags traffic in the Internet of Medical Things (IoMT), a label alone is not enough — the analyst must also know *which* attack family is at play and *why*. This paper introduces Clause-Activation Signatures (CAS), a method that turns the clauses of a trained weighted Tsetlin Machine (TM) into a compact, auditable signature for each attack family, so that attribution becomes signature matching rather than an opaque vote. Each signature is extracted with complementary-pairs stability selection (CPSS) over clause activations, split by coefficient sign into inculpatory and exculpatory evidence, and certified by a Shah–Samworth bound on spurious inclusions. Evaluated on CICIoMT2024 (WiFi/MQTT, family level) across three seeded TMs under a leak-free fit/selection/report split of the test set, CAS improves over the raw weighted-vote TM it is extracted from: macro-F1 rises from 0.692 ± 0.004 to 0.765 ± 0.029 , using only 28–53 sign-positive clauses per family from a 2,400-clause pool. Sorting clauses by weight and discarding the weakest 84% beforehand costs the base detector nothing, lifts CAS to 0.787 ± 0.023 , and removes its sensitivity to the stability threshold. The six signatures provably form a maximally 5-disjunct code on every seed, so any mixture of up to five families decodes exactly; under 5% bit-flip noise a thresholded decoder still recovers 99.8%–100%. A binary normal-versus-attack reduction lifts AUROC from 0.817 to 0.938. Two negative results are reported: weak open-set novelty detection under correct conformal false-flag control, and signature identity that does not survive retraining. CAS is a certified attribution layer that denoises a mediocre base detector, not a replacement for one.

Index Terms—attack attribution, disjunct codes, intrusion detection, interpretable machine learning, IoMT, open-set recognition, stability selection, Tsetlin Machine

I. INTRODUCTION AND RELATED WORK

An IoMT intrusion detector must both raise a correct alarm and name the attack family behind it in a form an analyst can audit, since containment and clinical risk differ by family. Post-hoc explainers such as SHAP, LIME and Anchors [14] serve only the first: they explain one prediction at a time with no stability guarantee. A per-family *signature* — interpretable

This work was supported by the *SecureIoTM—Ultra-low-energy IoT Intrusion Detection Systems using Logic-based Tsetlin Machines* project (Grant Number 342167), funded by the Research Council of Norway under the TEKNOLOGIKONVER programme. The authors acknowledge the support from the Centre of Artificial Intelligence Research at the University of Agder, Sørlandet Hospital Trust, and Newcastle University for their collaboration in this interdisciplinary research initiative.

Rakesh Reddy Yakkati, Lei Jiao, and Ole-Christoffer Granmo are with the Department of ICT, University of Agder, Grimstad 4879, Norway (emails rakeshry@uia.no, lei.jiao@uia.no, and ole.granmo@uia.no).

Per-Arne Andersen is with the Department of Information Systems, University of Agder, Kristiansand 4630 (email per.andersen@uia.no).

Rishad Shafik is with the Department of Microelectronic Systems, Newcastle University, Newcastle upon Tyne, UK (email rishad.shafik@newcastle.ac.uk).

by design, stability-vetted, attached to a family rather than a flow — is absent from the literature.

The Tsetlin Machine (TM) [1] is a natural substrate: its unit of computation is a conjunctive clause over Boolean literals that stays human-readable after training, and the weighted TM [2] adds an integer vote weight per clause. TM-based intrusion detection is established — rules on NSL-KDD [6], TMs on CICIoMT2024 and MedSec-25 [3], [7] — but these explain single predictions through whichever clauses fired, without error control. Clause-Activation Signatures (CAS) close that gap: per family, complementary-pairs stability selection (CPSS) [4], [8] over clause activations keeps the stably predictive clauses, splits them by coefficient sign into inculpatory and exculpatory evidence, and carries a Shah–Samworth certificate on spurious inclusions. Identifiability under simultaneous families follows from t -disjunct matrices [15]. We evaluate on CICIoMT2024 [12], the only peer-reviewed multi-protocol IoMT benchmark meeting standard validity criteria [9]–[11]; its schema carries no IP or port fields.

Contributions: a seed-reproducible CAS implementation on TMU [5] clause activations with a load-bearing sign filter (§II-F, §IV-A); three-seed $\pi_{thr}/B/\kappa$ sensitivity maps (§IV-C); certified per-family evidence lists (§IV-E); a clause-pruning program that discards 84% of the pool and improves CAS (§II-E, §IV-F); a maximal disjunct-code certificate (§IV-H); a binary reduction (§IV-D); and two negative results (§IV-G, §IV-I).

II. METHOD: CLAUSE-ACTIVATION SIGNATURES

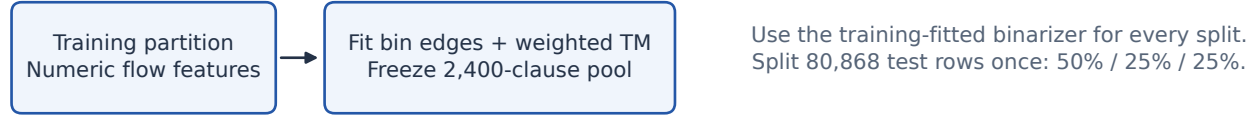
A. Overview

Fig. 1 shows the pipeline. Raw flows are first binarised with the standard thermometer binariser of the TMU library, and a weighted TM is trained on the resulting Boolean vectors. Everything afterwards works on the trained model’s learned rules: passing flows through the TM yields clause activations, and CPSS extracts a stable signature per class from them. Two guarantees accompany the signatures: a statistical error-control bound on spurious inclusions (§II-G) and a combinatorial identifiability guarantee for mixtures (§IV-H).

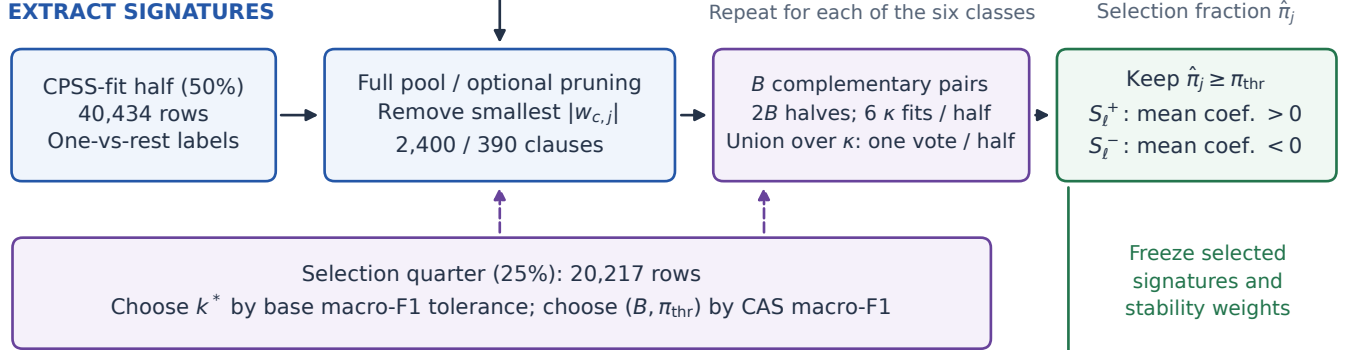
B. Dataset Description

We use CICIoMT2024 [12], WiFi/MQTT branch, with family-level labels. The dataset was collected on a testbed of 40 healthcare devices (25 real, 15 simulated) communicating over Wi-Fi, MQTT, and BLE. It contains 18 attack types grouped into five attack families, and each flow is described by 22 numeric CIC-extracted flow features. With

TRAIN ONCE



EXTRACT SIGNATURES



REPORT WITH FIXED CHOICES

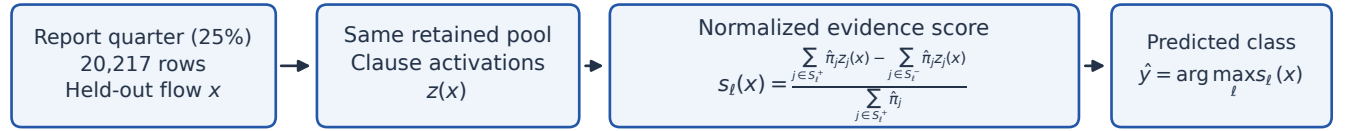


Fig. 1. CAS extraction and evaluation under the three-way protocol. The training partition determines the binarizer and weighted TM. Optional weight pruning precedes CPSS on the fit half; the selection quarter determines k^* , B , and π_{thr} (dashed arrows). Each complementary half contributes one selection vote per clause, using the union over the six-point κ grid. Both inculpatory S_{ℓ}^+ and exculpatory S_{ℓ}^- evidence must pass the stability threshold; mean coefficient sign separates them. The report quarter uses the fixed pool, signatures, and normalized score of (2). If the positive-weight denominator is zero, the implementation assigns $s_{\ell}(x) = 0$.

benign traffic there are six classes in all: Benign, distributed denial-of-service (DDoS), denial-of-service (DoS), attacks on the Message Queuing Telemetry Transport protocol (MQTT), reconnaissance (Recon), and Spoofing.

C. Booleanisation and Base Detector

Each numeric flow feature is discretised into 10 quantile bins. We use scikit-learn’s KBinsDiscretizer with bin edges fit on the training partition only. Features such as `Rate` contain $+\infty$ values from division by near-zero flow duration upstream; we clip them to the finite training maximum before binning, otherwise the quantile edges collapse to a single degenerate bin (§IV-A). The bins are encoded as cumulative thermometer bits of the form “feature $\geq$ bin k ”. TMU handles literal negation internally.

D. Weighted TM

We train a weighted multiclass TM on the six classes with $K = 400$ clauses per class, $T = 320$, $s = 8.0$, and 25 epochs. Passing flows through the trained TM yields the clause-activation matrix $Z \in \{0, 1\}^{n \times CK}$ with n flows and $CK = 2,400$ clauses. Here $Z[x, j] = 1$ if and only if clause j fires on input x .

E. Clause-Pool Pruning

Not every clause in the pool is worth carrying into signature extraction, and which ones to drop can be posed exactly.

Write $w_{c,j}$ for the learned weight of clause j of class c and $z_{c,j}(x) \in \{0, 1\}$ for its activation, so the class sum is $f_c(x) = \sum_j w_{c,j} z_{c,j}(x)$ and $\hat{y}(x) = \arg \max_c f_c(x)$. A keep-mask $m \in \{0, 1\}^{C \times K}$ with deletion sets $D_c = \{j : m_{c,j} = 0\}$ gives $f_c(x; m) = \sum_j m_{c,j} w_{c,j} z_{c,j}(x)$, and we seek the smallest pool that preserves accuracy:

$$\min_{m \in \{0,1\}^{C \times K}} \|m\|_0 \quad \text{s.t.} \quad F_1(\hat{y}(\cdot; m)) \geq F_1(\hat{y}(\cdot; 1)) - \varepsilon. \quad (1)$$

This is a search over 2^{CK} masks (2^{2400} here) whose constraint is non-decomposable, macro-F1 sitting behind an $\arg \max$. We therefore optimise the perturbation it induces, $R(D_c) := \sum_{j \in D_c} |w_{c,j}|$, which controls (1) through the decision margin.

Proposition 1 (Certificate). *Let $R = \max_c R(D_c)$ and let $\gamma(x) = f_{\hat{y}(x)}(x) - \max_{c \neq \hat{y}(x)} f_c(x)$ be the full-pool margin. If $\gamma(x) > 2R$ then $\hat{y}(x; m) = \hat{y}(x)$.*

Proof. Since $z_{c,j}(x) \in \{0, 1\}$, $|f_c(x) - f_c(x; m)| = |\sum_{j \in D_c} w_{c,j} z_{c,j}(x)| \leq R(D_c) \leq R$ for every c . Writing $a = \hat{y}(x)$, for any $c \neq a$, $f_a(x; m) - f_c(x; m) \geq (f_a(x) - R) - (f_c(x) + R) \geq \gamma(x) - 2R > 0$, so a remains the maximiser. $\square$

Proposition 2 (Exact greedy). *For a budget $|D_c| = k$, $\min_{D_c} R(D_c)$ is attained by the k clauses of smallest $|w_{c,j}|$.*

Proof. $R(D_c)$ is a sum of k non-negative terms selected without replacement. If D_c contains j while excluding j' with $|w_{c,j'}| < |w_{c,j}|$, swapping them strictly decreases R ; no such swap exists precisely when D_c holds the k smallest magnitudes. $\square$

Magnitude ranking is therefore optimal for the surrogate rather than a heuristic, and by Proposition 1 it carries a per-input guarantee, at cost $O(CK \log K)$ against the 2^{CK} of (1). Sweeping k in steps of five and taking the largest k that meets the constraint gives the operating budget used in §IV-F; CAS is then extracted from the surviving pool exactly as below, with nothing else changed.

F. CPSS Signature Extraction and Sign Filtering

CAS works entirely in clause-activation space, where a flow is represented by the rules that match it. We build one one-versus-rest problem per family ℓ . CPSS then repeats the following B times: split the fit half of the test set (fit, §III) into two complementary halves, and on each half fit an ℓ_1 -regularised logistic model from the columns of Z to the one-versus-rest label of family ℓ — $2B$ half-fits per family in all.

Two knobs control this stage: κ , the inverse-regularisation strength of the logistic model inside each half-fit, and π_{thr} , the stability threshold applied after all fits are done. A small κ forces most coefficients to exactly zero, so only a few clauses survive a single half-fit. We do not fix one κ but sweep it over a geometric grid of 6 points from 0.001 to 0.1. A clause counts as selected by a half-fit if any grid point selects it. This union-over-grid rule follows [4].

Each half-fit casts one vote per clause. The stability score $\hat{\pi}_j$ is the fraction of the $2B$ half-fits that select clause j . The signature keeps the clauses that win enough votes: $S_\ell = \{j : \hat{\pi}_j \geq \pi_{\text{thr}}\}$. The knobs are not independent: κ sets which clauses can collect votes at all, and π_{thr} then cuts the resulting vote distribution, so the signature size at a given threshold depends on the grid. We restrict κ to the range that keeps each half-fit sparse on this 2,400-clause pool. A wider grid would inflate q_Λ and make the certificate vacuous.

One further step is needed. CPSS selects on $|\text{coefficient}| > 0$, which is correct for the error-control bound but wrong for attribution: a stably negative coefficient is evidence *against* ℓ , not for it. We therefore track the mean signed coefficient $\overline{\text{coef}}_j$ of each clause across fits and split the signature into $S_\ell^+ = \{j \in S_\ell : \overline{\text{coef}}_j > 0\}$ and S_ℓ^- . The positive part is inculpatory evidence, the negative part exculpatory. The certificate still applies to the full sign-agnostic S_ℓ ; we state this scoping gap openly. §IV-A shows that this split is not cosmetic.

The attribution score is

$$s_\ell(x) = \frac{\sum_{j \in S_\ell^+} \hat{\pi}_j z_j(x) - \sum_{j \in S_\ell^-} \hat{\pi}_j z_j(x)}{\sum_{j \in S_\ell^+} \hat{\pi}_j}, \quad (2)$$

and the prediction is $\hat{y} = \arg \max_\ell s_\ell(x)$. Matches to inculpatory clauses add weight. Matches to exculpatory clauses subtract weight. Normalisation by the inculpatory weight is required. Without it scores scale with signature size instead of match quality.

G. Error-Control Certificate

With q_Λ estimated as $\text{mean}(\hat{\pi}) \cdot p$, the Shah–Samworth bound gives $E[V] \leq q_\Lambda^2 / ((2\pi_{\text{thr}} - 1)p)$ with $p = 2400$. We report these bounds as loose wherever they are loose (§IV-E): they certify that most of each signature is likely genuine, not that it is near-noise-free.

III. EXPERIMENTAL SETUP

Stratified caps are applied for tractability: at most 40,000 training and 15,000 test rows per family, Benign (19,291) and Spoofing (16,047) in full, giving 195,338 training and 80,868 test rows. The distribution is heavily imbalanced (DDoS and DoS about 95.7% of flows [12]), so we report macro-F1 and per-class metrics. The test set is permuted once and split three ways: CPSS half-fits use fit (40,434 rows), every free choice — the pruning budget k^* , the number of pairs B , and π_{thr} — is made on a disjoint selection quarter sel (20,217), and the report quarter rep (20,217) is read exactly once, at the end. No number we report was used to choose anything, which the closed-world critiques of IDS evaluation ask for [9]–[11]. Multi-seed results use three independently trained TMs (seeds 42, 7, 123) and fully seeded CPSS.

IV. RESULTS

A. Validation of the Construction

Three checks precede the headline. The infinite-value clipping of §II-C is essential: without it macro-F1 collapses from 0.72 to 0.54. The sign split (§II-F) is not cosmetic: the naive sign-agnostic scorer — a $\hat{\pi}$ -weighted sum over the whole selected support S_ℓ — collapses to macro-F1 below 0.01 on every seed, while the signed scorer of (2) reaches 0.765. And $\pi_{\text{thr}}=0.6$ is unstable as well as weak (0.687 ± 0.033 , one seed at 0.642, against 0.743 ± 0.016 at $\pi_{\text{thr}}=0.7$), so we do not hand-pick a threshold: the operating point is selected on sel and reported once on rep (§III).

B. Task A: Closed-Set Family Attribution

At the operating point selected on sel ($B=15$, $\pi_{\text{thr}}=0.8$) CAS beats its base detector on every seed (Table I); raising B further adds nothing (0.766 at $B=15$ against 0.751 at $B=25$ on sel). The gains land where the base is weakest — Benign recall rises from 0.26 to 0.79, Spoofing precision from 0.43 to 0.59 — while DDoS and DoS stay mutually confused under both methods, a documented dataset property [13].

TABLE I
TASK A: THREE-SEED MACRO-F1 ON THE REPORT QUARTER OF CICIO MT2024. THE OPERATING POINT (B , π_{thr}) IS SELECTED ON THE DISJOINT SELECTION QUARTER; THE PRUNED POOL IS §IV-F.

Method	(B , π_{thr})	macro-F1
Weighted-TM argmax (base)	—	0.692 ± 0.004
CAS, full pool	(15, 0.8)	0.765 ± 0.029
CAS, pruned pool	(10, 0.9)	0.787 ± 0.023

C. Sensitivity to π_{thr} , B , and κ

Fig. 2 sweeps both knobs on the report quarter after pruning to 390 clauses (65 per class). At $B=15$, varying π_{thr} from 0.5 to 0.9 leaves macro-F1 within 0.777–0.779 and accuracy within 0.792–0.794. At $\pi_{\text{thr}}=0.8$, varying B from 5 to 25 leaves macro-F1 within 0.776–0.778 and accuracy within 0.790–0.793. Both metrics remain above the pruned weighted-TM baselines (0.688 macro-F1 and 0.720 accuracy). These sensitivity slices retain the original fixed settings for comparison; the pruned-pool operating point selected on the selection quarter remains $(B, \pi_{\text{thr}}) = (10, 0.9)$. The separate κ sweep uses the full 2,400-clause pool (Table II): across $[10^{-3}, 4 \times 10^{-2}]$ the mean stays within 0.724–0.763, degrading only at $\kappa=0.1$. This justifies the union-over-grid rule: at the selected threshold the union (0.765) matches or exceeds every single- κ row (best 0.763).

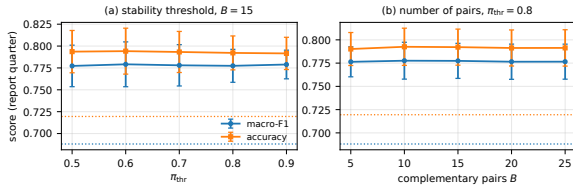


Fig. 2. Sensitivity after pruning to 390 clauses (65 per class), on the report quarter, three-seed mean $\pm$ sd. (a) CAS macro-F1 and accuracy versus π_{thr} at $B=15$; dotted lines are the pruned weighted-TM baselines. (b) versus the number of complementary pairs B at $\pi_{\text{thr}}=0.8$.

TABLE II

$\kappa \times \pi_{\text{thr}}$: THREE-SEED MEAN CAS MACRO-F1 ON THE REPORT QUARTER, $B=15$. THE BOTTOM ROW IS THE UNION-OVER-GRID RULE ACTUALLY USED, WHICH MATCHES OR EXCEEDS EVERY SINGLE κ AT THE SELECTED $\pi_{\text{thr}}=0.8$.

$\kappa \setminus \pi_{\text{thr}}$	0.5	0.6	0.7	0.8	0.9
1.0×10^{-3}	0.728	0.725	0.724	0.724	0.727
2.5×10^{-3}	0.748	0.745	0.739	0.741	0.746
6.3×10^{-3}	0.751	0.746	0.737	0.740	0.763
1.6×10^{-2}	0.748	0.742	0.752	0.752	0.745
4.0×10^{-2}	0.743	0.752	0.747	0.742	0.755
1.0×10^{-1}	0.673	0.693	0.718	0.722	0.694
union grid (method)	0.708	0.687	0.743	0.765	0.762

D. Binary Reduction: The Attack Signature

Collapsing the five attack families into one ATTACK class and training a dedicated two-class TM yields a binary CAS on the same booleanisation. The base TM over-predicts attack, catching almost everything (recall 0.987) but crying wolf often (precision 0.844). The attack signature ($B=25$, $\pi_{\text{thr}}=0.9$, selected on sel) rebalances the operating point (recall 0.910, precision 0.946) and ranks flows far better, lifting AUROC from 0.817 to 0.938 (Table III).

E. Signature Content and Certificates

At $\pi_{\text{thr}}=0.6$, $|S|$ ranges 115–311, $|S^+|$ 29–58, and $E[V]$ ≤ 72 – ≤ 250 . At the selected point $\pi_{\text{thr}}=0.8$ (Table IV, left)

TABLE III
BINARY ATTACK DETECTION ON THE REPORT QUARTER, THREE SEEDS (MEAN).

Method	attack F1	R / P	AUROC
Binary TM argmax	0.910 $\pm$ 0.000	0.987 / 0.844	0.817 $\pm$ 0.045
CAS signature	0.928 $\pm$ 0.008	0.910 / 0.946	0.938 $\pm$ 0.006

each family shrinks to 28–53 sign-positive clauses while F1 improves. The decoded DDoS signature is the classic flooding profile (Rate ≥ 31.5 k packets/s, short headers, ICMP tests), while exculpatory lists are long low-rate conjunctions resembling benign chatter.

F. Clause-Pool Pruning and CAS on the Pruned Pool

Solving the pruning program of §II-E, Fig. 3 traces macro-F1 on the selection quarter as k grows in steps of five: flat until 335 of the 400 clauses per class (84%) are gone, so $\varepsilon = 0.01$ gives $k^* = 335$ without touching any evaluation data. The surrogate certifies but does not solve (1): a global knapsack on $\sum_c R(D_c)$, or the minimax rule equalising $\max_c R(D_c)$, changes macro-F1 by less than the cross-seed spread (0.689, 0.678, 0.683 at 390 clauses).

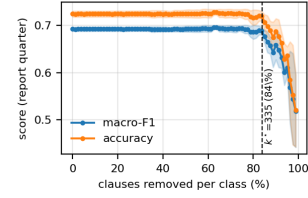


Fig. 3. Clause-weight pruning on the trained weighted TMs, three-seed mean $\pm$ sd on the report quarter. Clauses are removed per class in order of increasing |weight|, five at a time. Macro-F1 and accuracy are flat until 84% of the pool has been deleted.

If 84% of the pool is redundant for the base detector, is it also diluting the stability vote? Keeping only the top 65 clauses per class — the knee of Fig. 3 — and re-running the identical pipeline on the 390-clause pool answers yes (Fig. 4). Pruning costs the base detector nothing (0.688 ± 0.008 vs. 0.692 ± 0.004), but CAS reaches 0.787 ± 0.023 at its selected point ($B=10$, $\pi_{\text{thr}}=0.9$), above the 0.765 ± 0.029 from the full pool. More useful is the effect on the threshold: the full-pool sweep swings from 0.687 to 0.765 with π_{thr} , whereas the pruned pool is flat at 0.775–0.787 throughout, so the instability of §IV-A is a property of the diluted pool, not of CPSS. Signatures compact correspondingly (Table IV, right): cores of 22–42 clauses from a six-times-smaller pool, $E[V]$ tightening to ≤ 27 – ≤ 50 .

G. Open-Set Novelty

Leave-one-family-out with split-conformal thresholds ($\epsilon=0.05$, two folds) is negative: the false-flag guarantee holds (FPR 0.035 Spoofing, 0.027 MQTT, both under target) but power is weak (Spoofing AUROC 0.275, below chance;

TABLE IV

SIGNATURE COMPACTION AT $\pi_{\text{thr}}=0.8$ (SEED 42) ON THE FULL 2,400-CLAUSE POOL AND ON THE WEIGHT-PRUNED 390-CLAUSE POOL.

Family	full pool (2,400)		pruned (390)	
	$ S $	$ S^+ $	$ S $	$ S^+ $
Benign	184	41	63	30
DDoS	63	29	66	36
DoS	247	33	83	42
MQTT	145	53	64	26
Recon	66	28	58	22
Spoofing	194	50	76	28

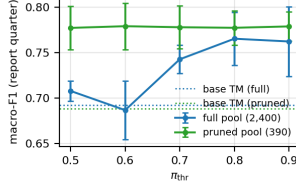


Fig. 4. CAS macro-F1 versus π_{thr} on the full and weight-pruned clause pools, three-seed mean $\pm$ sd on the report quarter at $B=15$. Pruning lifts the curve and flattens it: on the pruned pool the stability threshold no longer changes the result.

MQTT 0.622) — ARP-spoofing closely resembles legitimate ARP discovery in this feature space, a measured blind spot.

H. Mixture Identifiability

With the six inculpatory signatures as columns of a binary clause-by-family table, the largest t for which no column is covered by the union of any t others is $\hat{t}=5$ — the maximum for six families — on every seed and threshold, so any noiseless mixture of up to five families decodes exactly. Under 5% noise (500 trials) strict COMP collapses to 0.2%–1.8%, while a partial-match decoder restores 99.8%–100% at threshold 0.8 and 68%–76% at 15% noise.

I. Cross-Seed Signature Stability

The pairwise Kuncheva index on clause-index overlap across seeds is near zero for every family (−0.015 to 0.111): CAS stabilises selection *within* a trained TM — what attribution relies on — but signature identity does not survive retraining.

V. DISCUSSION AND CONCLUSION

CAS turns a trained TM’s clause pool into compact signed evidence lists per family, with a bound on spurious inclusions and exact mixture identifiability. On CICIoMT2024 it beats its base detector (macro-F1 0.765 vs. 0.692; binary AUROC 0.938), and discarding 84% of the pool by weight first raises this to 0.787 and removes the threshold sensitivity. All operating points are chosen on a disjoint selection quarter (§III), so no reported number was used to choose anything. Limitations: no post-hoc XAI baseline, loose $E[V]$ bounds, near-chance cross-seed identity, single-dataset scope.

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